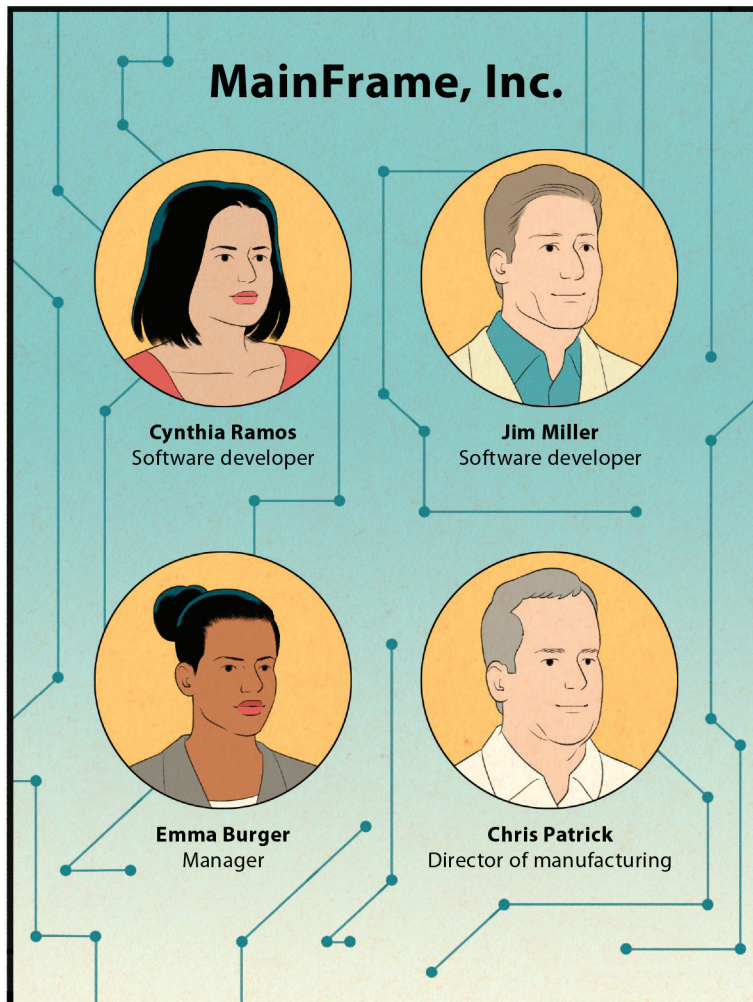




Case Study

Should I Pitch a New Project-Management System?

by Denis Dennehy

HBR's fictionalized case studies present problems faced by leaders in real companies and offer solutions from experts. This one is based on a case taught by Nien-hê Hsieh, the Kim B. Clark Professor of Business Administration at Harvard Business School.

CYNTHIA RAMOS JUST wanted to grab her pasta salad from the communal refrigerator and eat at her desk, but instead she got sucked into another exchange with Jim Miller. Since she'd joined MainFrame as a software developer, six months earlier, every conversation with Jim had been about what a hindrance the company's project-management system was. On her very first day—in a roomful of new colleagues—he'd asked whether her former employer had used Scrum.¹ When she'd said no, he'd chuckled and said, "You might regret leaving to come here!"

Now Jim shouted a greeting: "Hey, Ramos, how's *your* sprint going?" He and his tablemates—Adnan Persaud, and Wai Quon—burst into laughter. "Fine," Cynthia replied, smiling and grabbing her lunch. As she left the room, she heard Jim whisper, "I know she hates Scrum as much as I do."

Cynthia didn't love Scrum, but she did enjoy working at MainFrame. Her prior employer, Ambitious Computing, a start-up focused on artificial intelligence, was more technologically advanced, and her colleagues had been happier, but MainFrame was



Cynthia listens to Jim's pitch for transitioning to less-cumbersome processes and considers whether to help him.

I've been waiting for someone like you to join MainFrame. They'll listen to you.

one of the three leading hardware and software companies in the world. It had been in business for 40 years, and it paid her almost 50% more than Ambitious had.

The hardest change had been to go from working in a much quicker, more loosely structured manner to adjusting to MainFrame's more methodical and deliberate pace. Cynthia's nine-person team included developers in India, Australia, the UK, and the United States. They operated on a follow-the-sun model that maximized productivity by ensuring that work proceeded continuously, with

team members in one time zone handing off work to colleagues in another from day to day. Clear cross-team communication was essential.

Following the Scrum process,² each MainFrame software team's work was divided into two-week sprints with a major code release each quarter. The code would be tested and integrated with clients' real-time critical systems—such as military equipment, satellites, autonomous vehicles, and medical devices—that had to perform with zero failure.

At the end of each two-week sprint Cynthia would send a batch of code to a testing team and gather her own team to reflect on how things had gone and what they would like to change for the next sprint.³ She would also review the backlog—a prioritized list of projects that the testing team had identified for her team to work on next, such as features to support or bugs to fix. The list and the associated product road map were generated and approved by a management team high above her pay grade. Cynthia's team typically completed 10 stories (or tasks) during each sprint. MainFrame wanted everyone to complete at least 12. Despite his complaints, only Jim Miller ever hit that quota.

Cynthia, Jim, Adnan, Wai, and the other "Scrum masters" reported to Emma Burger, the manager responsible for gathering detailed user requirements from the business side and conveying them to the software teams. Emma reported to Chris Patrick, the director of manufacturing, who supervised 34 reports, including technical project managers, software development engineers, and release

managers in the United States and around the globe.

Cynthia enjoyed working for Emma and wanted to make her life easier, not harder. But she agreed with Jim: MainFrame's processes were too cumbersome. That's why, when Jim followed her out of the canteen to her cubicle, sat on her desk, and asked her to help him change things, she listened and ultimately agreed to meet with him later to discuss a plan.

JIM'S REQUEST

"I've been waiting for someone like you to join MainFrame," Jim said. "They'll listen to you."

"What do you want me to do?"

"Working in small batches, with these tight cadences, limited by management in what we're able to do, we're never going to be as productive as they want us to be," he said. "It takes so damn long to finally release even the smallest update. We need a different system. Have you used Flow? My daughter is a software developer too, and that's how her team works."

Not only had Cynthia used Flow analytical tools and techniques when she was at Ambitious Computing, but she'd earned a diploma in the method. She hadn't really wanted to take the course, but her boss and mentor at the time had insisted. "This is where the industry is headed," she'd said.

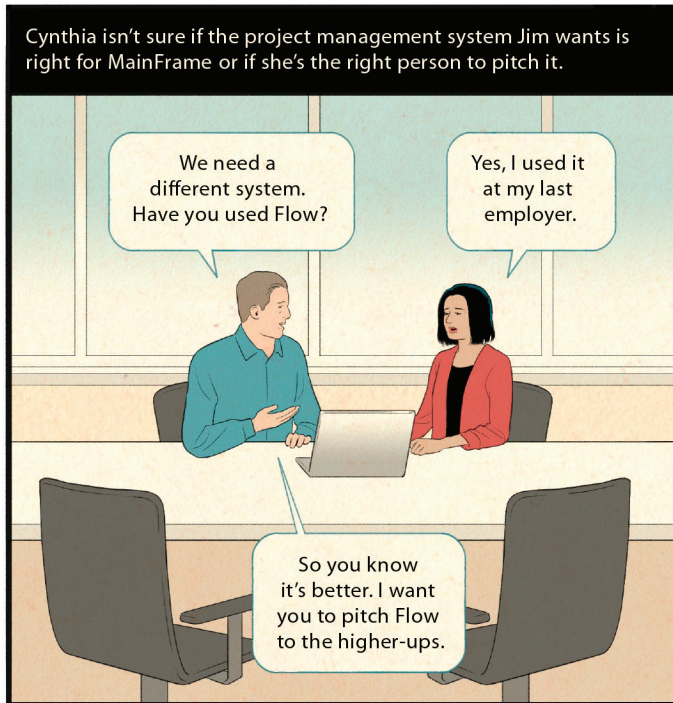


Case Study Classroom Notes

1. The term "scrum" (in a product development context) was introduced in Harvard Business Review ("The New New Product Development Game," HBR, January 1986).

2. Scrum was originally designed for software developers who work on cross-functional teams to deliver products and services in short cycles. The goals were fast feedback, quicker innovation, continuous improvement, and other benefits.

3. Flow and Scrum both derive from the methodology introduced in The Agile Manifesto, published in February 2001 by 17 software developers who wrote it during a ski vacation.



4. Scrum processes, most of which last two to four weeks, have clear start and end dates, whereas Flow projects are simply worked on until they're completed. What are the pros and cons of each approach?

5. Which aspects of Scrum or Flow would you introduce to your workplace? Which would you avoid? Why?

6. Should Cynthia trust Jim's intentions? Why or why not?

"I used Flow at Ambitious," Cynthia told Jim.

"So you *know* it's better," he said. "There's no continuous and unnecessary starting and stopping, no siloed workflows, more freedom.⁴ No more redundant daily 9 AM stand-up meetings that my team in Mumbai has to join at 6:30 PM their time. We set the release date. We leverage predictive analytics, using AI rather than asking for extended deadlines."

Cynthia agreed with Jim on those points, but she also knew that not everyone liked working with Flow. For example, the AI-based tools Jim referenced were great for predicting errors and impediments to the smooth flow of work, but that was only because all work was being monitored and managed transparently in real time. Teams working in Flow—and their managers—could see the development

process laid out on digital screens in their offices—which was maybe great for collaboration in distributed teams, but nerve-racking for less-confident developers.⁵

Furthermore, MainFrame was a technology stalwart, one of the first companies ever to produce commercial software. Cynthia wasn't sure that after only six months at a company that had been successful for 40 years she could ask its leaders to change the way they were doing business.

"How can I help you with this?" she asked.

"I want you to pitch Flow to the higher-ups," Jim said.⁶

A PILOT PROJECT

Against her better judgment, Cynthia agreed to attend a meeting with Emma and Jim. He did most of the talking, explaining his vision for what a pilot program

would look like and how he thought Cynthia could present it to Chris Patrick at the annual process-improvement review.

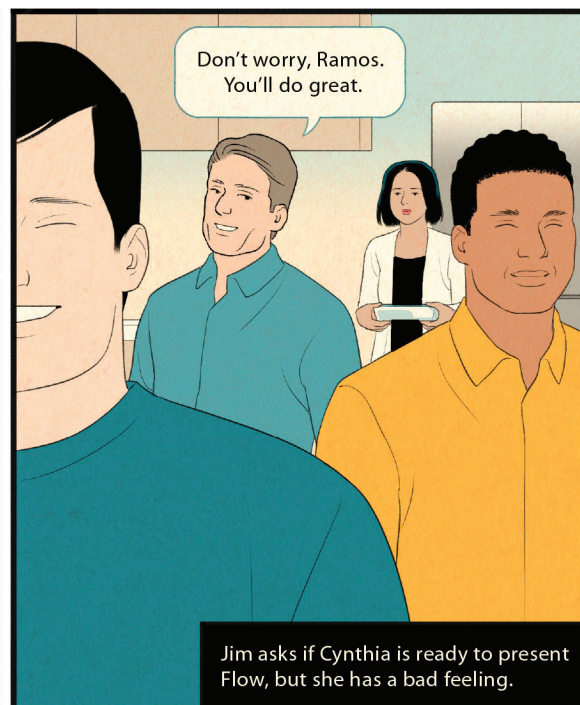
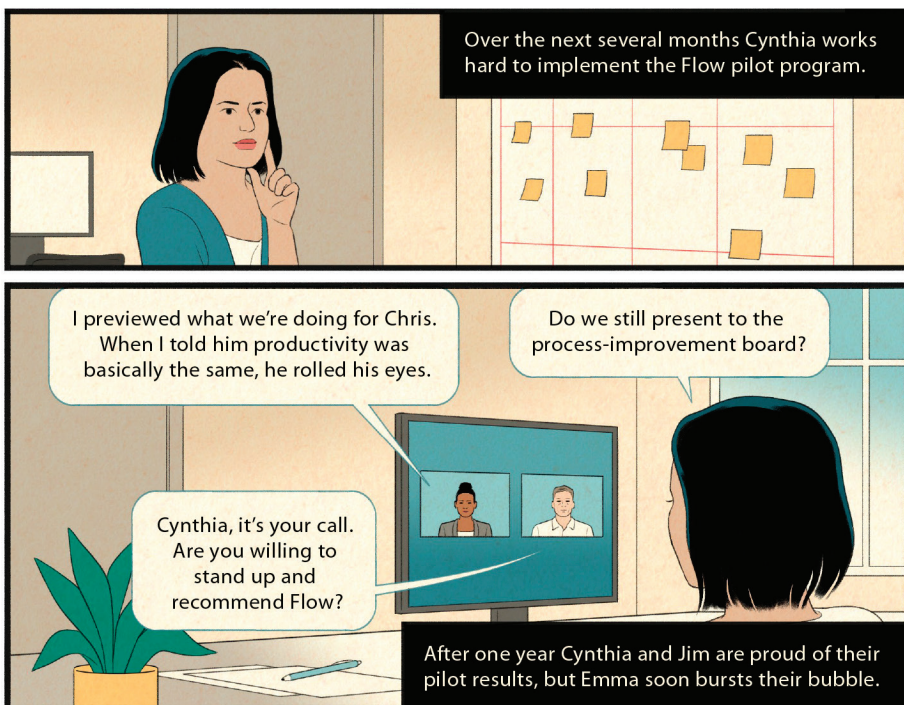
Emma was skeptical. "Cynthia, what do you think?"

"Flow worked very well at Ambitious," Cynthia said. "I'm not sure it's what we need here, but I'm happy to help pilot it."

"Would you be willing to present the results to the board?" Emma asked. Cynthia shrugged.

"Cynthia has an Ivy League degree," Jim said. "She worked for a future unicorn. She has a diploma in Flow. If anyone can demonstrate and communicate the need for this change to Chris, it's Cynthia."

Cynthia was taken aback by how much Jim knew about her but also pleased by his esteem. Since joining MainFrame, she'd kept her head down and tried her best to hit the unrealistic deadlines set for her by her



supervisors. Maybe it was time for her to speak up and take a risk.

“OK, as long as you all meet your quotas, I don’t care how you do it,” Emma said. “But if there are any signs of problems, I’m going to shut it down.”

Over the next few months Cynthia reviewed academic research papers and consultancy reports to compare techniques and challenges in Flow with those in Scrum and other software development methods such as Extreme Programming (XP) and feature-driven development. She reread *Stop Starting, Start Finishing* and similar books. She then told Emma she was ready to work with a vendor to launch a pilot project on its AI-powered Flow software, which, unlike their current system, had digital Kanban Board functionality for better team communication and could automatically generate metrics, such as cycle time and lead time,

which teams in Scrum needed to log and track themselves.

In sprint planning meetings, now scheduled for half a day rather than squeezed into an hour, Cynthia’s team discussed the product backlog, divided projects into tasks, and planned for future sprints to deliver on prioritized requirements. In some instances Cynthia had to devise a combination of Scrum and Flow techniques, especially at the beginning of projects.⁷ The frequent-delivery approach of two-week sprints was a Main-Frame requirement. That rule couldn’t be changed until everyone was working in Flow.

The pilot was a hot topic in the canteen. On the day Cynthia announced that her team would be experimenting with Flow coordination, Wai spoke to her directly for the first time. “Thanks for doing this!” he said. “I can’t wait until we all get it!”

ONE YEAR LATER

After the pilot had run for a year, Cynthia was proud of the results. Her team had maintained its 10-story-per-sprint production but was now continuously deploying code with no critical errors, and team morale had improved. Jim had cheered her on the whole way.

“You really did it,” he told her. “We’ve got AI choosing and prioritizing your projects rather than some corporate nitwit who has never written a line of code. It’s catching defects and bottlenecks before your people do. The bosses can see exactly where the team is in its projects and exactly how little time they’re giving you!”⁸

But Emma soon burst their bubble. “I previewed what we’re doing for Chris,” she said on a video call with Cynthia and Jim from London, where she’d traveled for a senior planning



7. The five Scrum values are commitment, focus, openness, respect, and courage. Which of those values has Cynthia demonstrated?

8. How will the introduction of Flow change the way employees and managers interact with one another?



meeting. “We got into a big argument. He’s never used Flow. The first thing he asked was how many stories Cynthia’s team averaged each sprint. When I told him productivity basically stayed the same, he rolled his eyes.”

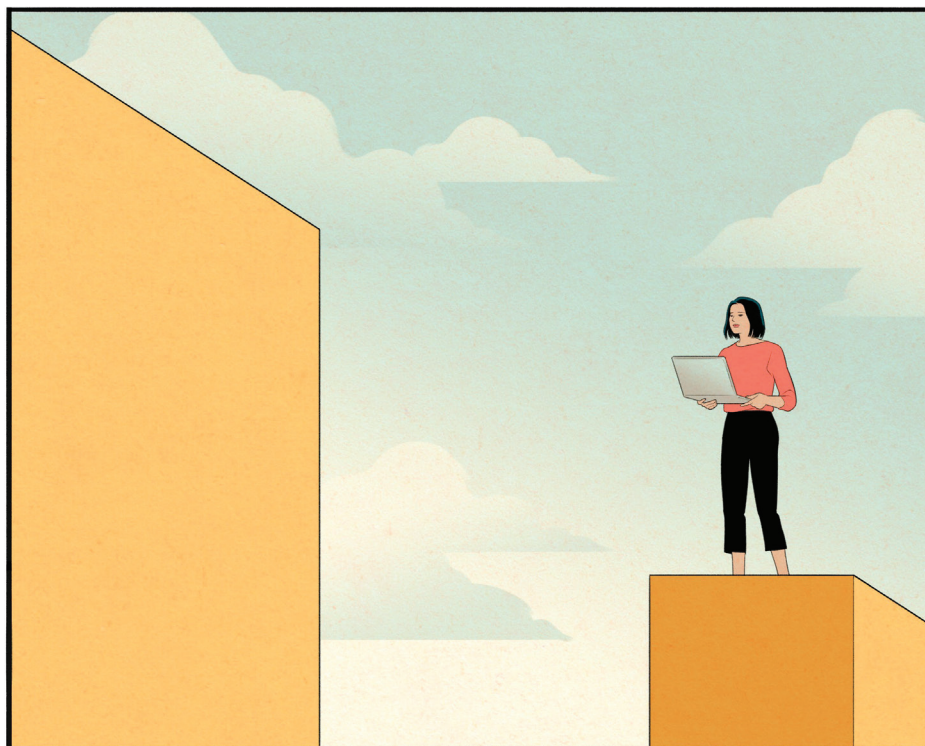
“So is that it?” Cynthia asked. “Do we still present to the process-improvement board? Can we keep the pilot going?”

“Chris said it was our decision, but he’s confident that we’re going to get roasted by the committee—especially the older members, who don’t have a software background. Honestly, Cynthia, you have the expertise, and you ran the pilot, so it’s your call. Do you believe this will improve our productivity in the long run? And are you strong enough in that conviction to stand up in front of these guys and say so?”

That night, as she was leaving the office, Cynthia ducked into the canteen to grab her unopened Tupperware container. She hadn’t been hungry for lunch that afternoon. Her stomach had been tied up in knots. Wai, Adnan, and Jim were leaving the office for happy hour and asked if she was ready to present Flow. She said she had a bad feeling.

“Don’t worry, Ramos,” Jim said. “You’ll do great.” ☺

 **DENIS DENNEHY** is an associate professor of business analytics and the school research lead at Swansea University’s School of Management, in Wales.



Should Cynthia present Flow to the process-improvement board?

The experts respond.



SONALI RAUT is a senior data scientist at Munich RE Automation Solutions.

Cynthia should wait to present Flow to the process-improvement board.

First she should meticulously craft a proposal delineating the rationale behind this initiative, including well-researched data and evidence-based arguments. That will improve the

prospects of successfully implementing the methodology.

Her biggest error was failing to include Chris Patrick on the Flow project from the start. As an established company with a track record of success, MainFrame won’t introduce a new project-management system without his support.

But make no mistake: Adding Flow to the company’s processes would absolutely improve project structure, adaptability, efficiency, administrative overhead, and productivity. It’s Cynthia’s responsibility to bolster her proposal with data, research findings, and pertinent case studies.



The Scrum Guide says that people who use Scrum methods should be comfortable enough with colleagues to speak up for what they believe in.

Now that Chris finally knows what Cynthia and Emma are doing, they should walk him through everything the team has accomplished. Knowing that he's focused on stories-per-sprint as a key metric, and that her team failed to improve on that during the pilot, Cynthia should highlight the transparency provided by the digital Kanban board, the improved teamwork, and the higher morale.

She should emphasize that an unchanged story-per-sprint metric is actually a positive indicator: Her team was able to sustain productivity during the pilot even as it transitioned to continuous code integration and deployment, improved oversight, and made developers happy. As the team becomes more adept at the new system, speed and efficiency will only increase.

Cynthia must strive to make Chris understand that other metrics—such as enhanced product features and reduced development costs—matter too. Perhaps she can offer an example of an improvement made during the pilot that delighted customers, or an error caught by developers that they would have missed without using Flow. She should also show Chris case studies from other companies that have transitioned to Flow and explain how her previous employer uses it.

Finally, she will have to detail the business case for switching. Does she know how much the transition will cost? Does she have a budget in mind for the project? How long will it take? How much money will the company gain or save by improving productivity, collaboration, and transparency? That is, what's the ROI?

Only after Chris is on board, and she, he, and Emma have settled these issues, should she make her presentation to the process-improvement board. I'm confident that Cynthia can persuade MainFrame to switch to Flow from Scrum, but she's not ready to make her case yet.



ALEX ESTEVAM is a technical program manager at Mastercard.

Cynthia is ready to present.

One of the five key values of Scrum is courage. *The Scrum Guide*, which was written and amended by the originators of the Scrum framework, says that people who use Scrum methods should be comfortable enough with colleagues to speak up for what they believe in. That's what Cynthia must do here.

She has good reason for wanting to introduce Flow processes to MainFrame. Scrum methods require a top-down approach. Senior management chooses what developers work on and tells them how long they have to work on it, which developers don't like. Flow, by contrast, gives developers the ability to show management what needs to be fixed and when. Also, many of her colleagues want Flow—and Cynthia seems to as well. So she should present it.

Of course, no matter how effectively she presents Flow to the board, some managers and even some developers will resist it. Although I know some

developers who absolutely hate those daily stand-up Scrum meetings—which they regard as a waste of time and believe that managers conduct only because they have to—others think they're essential, allowing everyone to collectively prepare for the day ahead.

Nonetheless, Cynthia knows that a transition to Flow is the right long-term move for MainFrame. That's why it's critical for her to present to the board. Whether she succeeds will probably depend on how well she educates the board members. She has already proved that her team can maintain productivity, continuously rolling out new code with no major errors. That's a victory.

She should also find a financial justification for the switch. She should make it clear that she can expand the pilot and start collecting data to demonstrate that the processes have improved and that those improvements will ultimately save MainFrame a lot of money.

If no one has the courage to present Flow, management may never learn enough about it to understand how well it works. Cynthia's colleague is right. She's the person best placed to persuade skeptics like Chris.

One other thing I want to note: *The Scrum Guide* says courage should be bilateral. That means that if Cynthia is brave enough to present Flow to the board, Chris must have the courage to listen to her argument and, if she makes a good one, admit that he was wrong about Flow. ☺

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